Spring End semester 2025 question paper					
Spring 2025 :: Offered Courses (DPS)					
				Submitted	Received
Sl. No.	Course Co	Course Name	Instructor(s)		
1	PH1201	Electricity and Magnetism	Ananda Dasgupta,	✓	
			Siddhartha Lal		
2	PH2202	Thermal Physics	Koushik Dutta	✓	
3	PH2204	Mathematical Methods of Physics I	Pradeep Kumar Mohanty	✓	
4	PH3201	Basic Statistical Mechanics	Anandamohan Ghosh	✓	
5	PH3202	Intermediate Electricity and Magnetism	Rajesh Kumble Nayak	✓	
6	PH3203	Advanced Quantum Mechanics	Sourin Das	✓	
7	PH4202	Advanced Statistical Mechanics	Bhavtosh Bansal	✓	
8	PH4203	Research Methodology	Soumitro Banerjee	✓	
9	PH4204	High Energy Physics	Ritesh Kumar Singh	✓	
10	PH4205	General Theory of Relativity and Cosmology	Golam Mortuza Hossain	✓	
11	PH4207	Quantum Information Processing	Chiranjib Mitra	✓	
12	PH4209	Evolutionary Dynamics	Supratim Sengupta	✓	
13	PH4213	Symmetry Methods in Physics	Ananda Dasgupta	✓	

Symmetry Methods in Physics

PH4213 End Semester Examination 2025

Total Marks : 50 | Time : 150 minutes

Answer Q 5 and any two questions from the rest

To help jog your memory,

- A subgroup N of G is a normal subgroup if $\forall n \in N$ and $\forall g \in G$, $gng^{-1} \in N$.
- The GOT says
$$\sum_{g\in G} D^{(\alpha)}_{ii'}\left(g^{-1}\right) D^{(\beta)}_{j'j}\left(g\right) = \frac{|G|}{d_\alpha}\delta^{\alpha\beta}\delta_{ij}\delta_{i'j'}.$$
- For a Lie group with composition functions $f^j\left(\alpha;\beta\right)$ we have
$$\left[H\left(\alpha\right)\right]^j{}_k \equiv \eta^j{}_k\left(\alpha\right) \equiv \left.\frac{\partial f^j}{\partial \beta^k}\left(\beta;\alpha\right)\right|_{\beta=0}$$
and the $\xi^j{}_k\left(\alpha\right)$ are components of the matrix $\Xi\left(\alpha\right) = H^{-1}\left(\alpha\right)$.
- For small α, β the composition function has the form: $f^j\left(\alpha;\beta\right) = \alpha^j + \beta^j + a^j_{kl}\alpha^k\beta^l + \ldots$
- The structure constants obey: $C^j_{kl} = a^j_{kl} - a^j_{lk}$.
- The group element that is attained for $\sigma = 1$ for an OPS with initial tangent vector t is formally denoted $\exp t$.

Q 1a) Starting from the GOT prove that a representation of a finite group G is irreducible if and only if its character obeys

$$\sum_{g\in G}\left|\chi\left(g\right)\right|^2 = |G|$$

You will have to prove all the intermediate results that you are using in the process.

b) A particular group has two 1 element classes, six 5 element classes and two 4 element classes. Deduce the dimensionalities of its irreducible representations. *You don't have to prove the equation(s) and result(s) that lead you to the answer - but you do have to justify why your answer is the only possibility.* [7 + 5]

Q 2) a) Find all normal subgroups of the cyclic group of order 6 formed by the elements

$$\{r, r^2, r^3, r^4, r^5, r^6 = e\}$$

b) Consider the subset T of $SL(2, \mathbb{R})$ defined as follows

$$T = \left\{ \begin{pmatrix} x & y \\ 0 & x^{-1} \end{pmatrix} : x, y \in \mathbb{R},\ x \neq 0 \right\}$$

Is T a normal subgroup of $SL(2, \mathbb{R})$?

c) The group S_4 of all permutations of 4 objects has 24 elements.

i) How many classes does S_4 have?

ii) Determine the number of elements in each of these classes of S_4.

Hint : remember the connections between classes of S_n and the cycle-structure of permutations.
$[3 + 4 + (2 + 3)]$

Q 3) a) Show that the set of all 2×2 matrices with integer entries and unit determinant, $SL(2, \mathbb{Z})$ form a group under matrix multiplication.

b) Consider two specific elements of $SL(2, \mathbb{Z})$:

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

i) Show that $S^{2n} = (-1)^n \mathcal{I}$ and $S^{2n+1} = (-1)^n S$. Hence Calculate $\exp S$.

ii) Show that the set $\{T, T^2, T^3, \ldots\}$ form an infinite cyclic group.

c) It is obvious from (b-ii) above that S generates a cyclic subgroup of order 4. Find the order of the other finite cyclic subgroups of $SL(2, \mathbb{Z})$. *Hint: if $A \in SL(2, \mathbb{Z})$ is the generator of a finite group, then its eigenvalues must be roots of the unity.* $[4 + (2 + 2) + 4]$

~~Q 4) a) Find a 2D matrix representation for the symmetries of a square~~. You can choose any basis you want - but a smart choice will make your life a lot simpler.

b) Consider a representation of the group S_3 with the reduction $2A_1 \oplus 3A_2 \oplus 3E$ (A_1, A_2 and E are 1,1, and 2 dimensional irreps, respectively, with A_1 being the identity representation). Comment on the structure of a matrix M which commutes with all the group representation matrices. Deduce the degeneracies of the spectrum of M. $[8 + 4]$

Q 5) Consider the set formed by matrices of the form $\begin{pmatrix} 1 & \alpha^1 & \alpha^3 \\ 0 & 1 & \alpha^2 \\ 0 & 0 & 1 \end{pmatrix}$, $(\alpha^1, \alpha^2, \alpha^3) \in \mathbb{R}^3$. *Be careful about the position of the parameters within the matrix.*

a) Show that this set forms a group under matrix multiplication. [3]

b) Using $(\alpha^1, \alpha^2, \alpha^3)$ as a coordinate for the matrix $\begin{pmatrix} 1 & \alpha^1 & \alpha^3 \\ 0 & 1 & \alpha^2 \\ 0 & 0 & 1 \end{pmatrix}$ determine the composition functions $f^j(\alpha; \beta)$. [2]

c) Show that this group is a Lie group. [4]

d) Show that

$$H(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \alpha^2 & 0 & 1 \end{pmatrix}$$

and hence determine $\Xi(\alpha)$, for these coordinates. [4]

e) Verify that

$$\frac{\partial f^j}{\partial \alpha^k}(\alpha; \beta) = \eta^j{}_l (f(\alpha; \beta))\, \xi^l{}_k(\alpha)$$

is valid here (*it may be better to use the matrices H and Ξ*). [3]

f) Determine the structure constants with respect to these coordinates. [2]

g) Write down that the operators $X_k \equiv \eta^j{}_k(\alpha) \frac{\partial}{\partial \alpha^j}$ explicitly. Verify that they obey the commutation relations $[X_i, X_j] = -C^k_{ij} X_k$. [3]

h) Solve the equations for the coordinates of the one-parameter subgroups

$$\frac{d\alpha^j}{d\sigma} = \eta^j{}_k(\alpha(\sigma))\, t^k$$

for any given tangent vector (t^1, t^2, t^3), and hence determine the group element (the matrix) $\exp t$. [3]

i) Show by explicit matrix multiplication that $\exp(\sigma t) \exp(\tau t) = \exp((\sigma + \tau) t)$ [2]

Ananda DasGupta

Evolutionary Dynamics (PH4209) Final Exam (April 30, 2025)

Answer all questions. **Total Time=2.5 hours. Total Marks=50**. The marks are specified in brackets.

1.a) The following reactions lead to Rock-Paper-Scissors dynamics. Write down the set of dynamical equations for the frequencies ($x_A; x_B; x_C$) of A, B, C respectively. ϕ – empty site; p,q,z are the rates of the respective reactions (3)

$$AB \xrightarrow{p} A\phi, \quad BC \xrightarrow{p} B\phi, \quad CA \xrightarrow{p} C\phi,$$
$$AB \xrightarrow{z} AA, \quad BC \xrightarrow{z} BB, \quad CA \xrightarrow{z} CC,$$
$$A\phi \xrightarrow{q} AA, \quad B\phi \xrightarrow{q} BB, \quad C\phi \xrightarrow{q} CC,$$

b) If in addition to the above reactions, A, B, C can mutate to each other with a rate μ i.e. (3)

$$A \xrightarrow{\mu} \begin{cases} B \\ C \end{cases}, \quad B \xrightarrow{\mu} \begin{cases} A \\ C \end{cases}, \quad C \xrightarrow{\mu} \begin{cases} A \\ B \end{cases}$$

Write down the new set of dynamical equations for the frequencies $x_A; x_B; x_C$

2. In a game with two strategies A and B and a population of N players among whom **i** play strategy A and **(N-i)** play strategy B. The pay-off matrix elements E(A,A)=a, E(A,B)=b, E(B,A)=c, E(B,B)=d

a) Write down the total expected payoffs (F_i and G_i) to A and B respectively in a state where there are *i* A's and *(N-i)* B's. (2)

b) If evolution occurs according to a Moran process, write down the transition probabilities (α_i and β_i) for the number of A's to (i) increase by one (ii) decrease by one? (4)

c) If the invasion probability of A is given by $\rho_A = \frac{1}{1+\sum_{k=1}^{N-1}\prod_{i=1}^{k}\gamma_i}$ and the invasion probability of B is given by $\rho_B = \frac{\prod_{i=1}^{N-1}\gamma_i}{1+\sum_{k=1}^{N-1}\prod_{i=1}^{k}\gamma_i}$ and ***w*** is the strength of selection, derive an expression for ρ_B/ρ_A in the weak selection limit (***w<<1***) to first order in ***w***. (6)

d) Obtain the condition for strategy A to be **Risk Dominant** over strategy B in the weak selection limit in terms of N,a,b,c,d. Show how this condition leads to a constraint on the mixed state equilibrium frequency in the N>>1 limit. (4)

3. Consider a system with two types of agents A and B. If t_j is the *unconditional* average fixation time, starting from a state where #A's=j.

Write down the recursive equation satisfied by t_j. Use the equation to derive an expression for t_1 in terms of the invasion probability (φ_1) of A and the transition probabilities T_m^+, T_m^- (2+5)

4. (a) Imagine a structured population of size N described by a **cycle** graph where a node can be replaced by the offspring of a node preceding or following it. Write down the N x N stochastic matrix that specifies the probability with each node can replace any other node in the graph. (2)

(b) In the **cycle** graph, if the #B members are ***i*** and #A=***N-i***, what is the probability that the #B individuals (i) decrease by 1 and (ii) increase by 1? Justify your answer in each case. Calculate the

probability of fixation of a B mutant that arises in such a structured population. Assume fitness of B=r, fitness of A=1. You can use the fixation probability formula given in question 2. (2+3)

(c) (i) Consider a PD game (with pay-off matrix elements E(A,A)=a, E(A,B)=b, E(B,A)=c, E(B,B)=d) on a regular network, where each node has k-connections. $q_{A|B}$ represents the probability that a B-player has a A-player as a neighbour etc. If a A-player is selected for death through a death-birth process of evolution

Write down the payoff P_i to the ith-player (i=A-neighbour of the A player selected for death or B-neighbour of the A-player selected for death); explain each term in the expressions for the two payoffs. (2)

(ii) If a A-player is selected for death in the network, obtain an expression for the probability that the A-player is replaced by one of its B-neighbours. Explain the significance of each term in the expression. (4)

5. In a PGG game of **n** players, **c** is the cost of cooperating, **r** is the synergy factor, and the number of cooperators is **k**.

(a) If **r** is a constant, obtain the condition for which cooperation can be favoured even when defection is the optimal strategy (i.e. payoff to a D is larger than the payoff to C when #cooperators=**k<n**). (2)

(b) For non-linear PGG's when the synergy factor **r(k)** is no longer a constant but an arbitrary function of **k**, obtain the condition for which cooperation can be favoured despite defection being the more optimal strategy for **k<n**. (2)

(c) If $r(k) = r_1 + \frac{k-1}{n-1}(r_n - r_1)$ where $r_1 \equiv r(k=1) < r_n \equiv r(k=n)$, obtain the expression threshold value of cooperators $\boldsymbol{k_*}$ beyond which cooperation by a focal player leads to higher payoff than defection when there are $\boldsymbol{k}$ other cooperators. (2)

(d) Prove that FULL cooperation is rational when $k_* < (n-1)$ (2)

Quantum Information Processing.

Ph 4207, End Semester Exam 2025 (Total Marks 30)
(28/04/25)

1. A single qubit is represented by the point (0.6,-0.48,0.64) on the Bloch Sphere.

   (a) Is this a pure state? If it is, find the ket vector representing it.

   (b) What is the density matrix for this state?

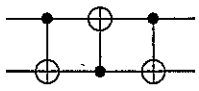
   (c) Calculate the $(\Delta X)^2$ for this state.

   (d) Find the point on the Bloch Sphere that is obtained by acting on this state by the **X** gate.

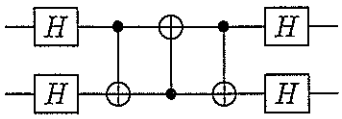
[4]

2. Figure out the action of the two quantum circuits below on the bell state $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$:

   (a)



   (b)



[2+2]

3. Consider the two qubit state

$$\rho = p\,|\psi\rangle\langle\psi| + \frac{1-p}{4}\mathbf{I}$$

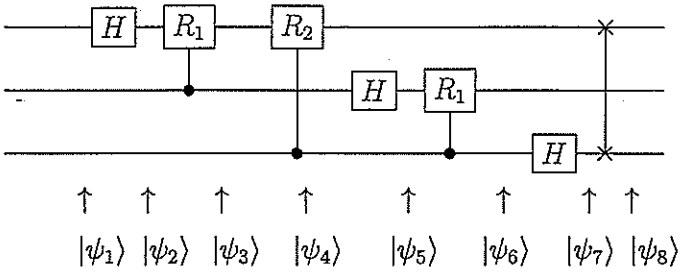
where $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Here $\boldsymbol{I}$ is the density operator on the 2-qubit Hilbert space.

   (a) Using the Peres-Horodecki criteria of partial transposition examine the entanglement of this state. Find the range of values of p for which the state is entangled.

(b) Again examine the entanglement of this state using Concurrence. Here too Find the range of values of p for which the state is entangled.

[4+6]

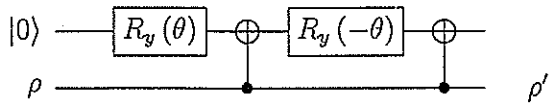
4. Consider a 3 qubit Gate, $|\psi_1\rangle = |101\rangle$, find each of the quantum states given in the Q-circuit below.



Where, H is the Hadamard Gate and $R_k = \begin{bmatrix} 1 & 0 \\ 0 & exp(2\pi i/2^k) \end{bmatrix}$

[6]

5. Consider the Kraus operators $F_0 = \begin{bmatrix} 1 & 0 \\ 0 & cos\theta \end{bmatrix}$ & $F_1 = \begin{bmatrix} 0 & 0 \\ 0 & sin\theta \end{bmatrix}$ Now consider the quantum circuit shown in the figure below



Show that this circuit induces a quantum operation $\rho \rightarrow \rho' = F_0 \rho F_0^\dagger + F_1 \rho F_1^\dagger$

[6]

Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH4205 :: General Theory of Relativity and Cosmology
End-semester Examination (Spring 2025)
Marks: 50, Duration: 2 hour and 30 min

1. Show that the 4-velocity $u^\mu = \frac{dx^\mu}{d\tau}$ of a particle is a contravariant vector under general coordinate transformations, where τ is the proper time. Further, show that the squared magnitude of the vector u^μ in an arbitrary spacetime with metric tensor $g_{\mu\nu}$ is $-c^2$ and that the metric tensor $g_{\mu\nu}$ is covariantly conserved, *i.e.*, $\nabla_\alpha g_{\mu\nu} = 0$. [3+3+3]

2. Compute all Christoffel symbols and express the equation for great circles in (θ, ϕ) coordinates to show that the geodesics on a unit two-sphere with the metric $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$ are great circles. [4+3+4]

3. A particle of mass m is moving along a geodesic in a spacetime spanned by the coordinates (t, r, θ, ϕ). The non-vanishing Christoffel symbols, of the form $\Gamma^\theta_{\mu\nu}$ and $\Gamma^\phi_{\mu\nu}$, are $\Gamma^\theta_{r\theta} = \Gamma^\phi_{r\phi} = \frac{1}{r}$, $\Gamma^\theta_{\phi\phi} = -\frac{1}{2}\sin 2\theta$ and $\Gamma^\phi_{\theta\phi} = \cot\theta$. Find the associated constant of motion for the particle. [5]

4. Compute the stress-energy tensor $T_{\mu\nu}$ for a scalar field Φ described by the action $S_\Phi = \int d^4x\sqrt{-g}[-\frac{1}{2}g^{\mu\nu}\nabla_\mu\Phi\nabla_\nu\Phi]$. By treating the stress-energy tensor as a perfect fluid, compute the corresponding equation of state. Given $\delta g = g g^{\mu\nu}\delta g_{\mu\nu}$. [4+3]

5. If the stress-energy tensor $T_{\mu\nu}$ of a perfect fluid is traceless, show that the corresponding equation of state is given by $w = \frac{1}{3}$. Solve the Friedman equation for a flat FRW universe containing the given perfect fluid. Plot the scale factor $a(t)$ and the energy densities $\rho(t)$ and $\rho(a)$. [3+4+3]

6. Write down the Killing equation for the metric $ds^2 = -c^2dt^2 + dy^2$. Find the general solution of the corresponding Killing equation. Show that there are three independent Killing vectors and write them down explicitly. [1+3+4]

G.M. Hossain

7/5/25

Date : 7th Mar, 2025 End-Sem

PH4204 : High Energy Physics

Duration: 2h30m (or 150 minutes) Total marks: 40

Time management tip: One marks is about 3 minutes (3.5 min to be exact).
First try the problem you are most comfortable with.
Useful relations are given at the back of the question paper.

1. **Pion decay:** The lightest particle with strong interaction, named $\pi^{\pm}$, decays weakly to a charged (anti-)lepton and corresponding (anti-)neutrino:

$$\pi^+ \to l^+ \, \nu_l, \qquad \pi^- \to l^- \, \bar{\nu}_l, \qquad l = e, \, \mu \, .$$

   (a) Assuming the decay amplitude to be same for the two channels, find out which channel is expected to be more likely and by what factor. Use $m_\mu/m_e = 200$.
   Hint: The phase space argument. 2

   (b) Given that branching ratio into μ is about 10^4 times more than the electron channel, systematically establish the nature of $\pi - l - \nu$ interaction vertes. Discuss all the possibility for the interaction vertex.
   Hint: Use Occam's razor after discussion all possibilities. 6

2. **Gauge interactions:** Consider a theory having $SU(3) \times SU(2) \times U(1)$ gauge symmetry.

   (a) Write down the Gauge field part of the Lagrangian density. Write the expressions for the field strength tensors for each of the gauge fields. Give the enumerated list of gauge bosons and list with which other gague bosons they interact. 3

   (b) The theory has a Dirac fields Ψ_1 which is a triplet under $SU(3)$, singlet under $SU(2)$ and has $U(1)$ charge to be $+1$. Write down the terms in the Lagragian density for Ψ_1 field. Expand the expression for D_μ operator here. 3

   (c) Consider another Dirac field Ψ_2 which is a singlet under $SU(3)$, doublet under $SU(2)$ and has $U(1)$ charge to be $+1$. Write down the terms in the Lagragian density for Ψ_2 field. Expand the expression for D_μ operator here. 3

   (d) Consider the Yukawa terms $\bar{\Psi}_1\Psi_1\Phi_1$ and $\bar{\Psi}_2\Psi_2\Phi_1$, where Φ_1 is a scalar. What are the possible quantum numbers of the field Φ_1 that allows both the Yukawa terms. Write down the gauge invariant kinetic term for Φ_1.
   Hint: Use tensor product decomposition like, $2 \otimes 2 = 3 \oplus 1$ *in* $SU(2)$. 3

   (e) What would be the quantum number of Φ_1 if $\bar{\Psi}_1\Psi_1\Phi_1$ type Yukawa terms is allowed but not $\bar{\Psi}_2\Psi_2\Phi_1$? Write down the expanded the expression for $D_\mu\Phi_1$ operator here. 3

   (f) Conside the Yukawa term $\bar{\Psi}_1\Psi_2\Phi_2$, where Φ_2 is another scalar field. Obtain the quantum numbers of Φ_2 that allows for this Yukawa term. Write down the gauge invariant kinetic term for Φ_2. 3

   (g) For Φ_1 in part (d) above and Φ_2 in the part (f) above, write down all possible 3- and 4-point interaction terms allowed by the gauge symmetries. 3

3. **The Standard Model and $e^+e^- \to \mu^+\mu^-$:** In the Standard Model the vector bosons Z and γ interact with both electrons and muons.

   (a) Knowing that γ interacts vectorially while Z interacts with general vector and axial-vector couplings, write down the interaction vertices. 1

   (b) For the process $e^-(k_1) + e^+(k_2) \to \mu^-(p_1) + \mu^+(p_2)$, draw the Feynman diagrams and write down the matrix element. 2

   (c) Obtain the expression for matrix element squared summed over spin for each diagram separately. Compute the traces and symplify the expressions using symmetry arguments. 4

   (d) Obtain the expression for interference term(s), sum over spins, compute traces and symplify the expressions using symmetry arguments. 4

Useful relations:

$\mathrm{Tr}\left[\gamma^\mu\gamma^\nu\right] = 4\eta^{\mu\nu}$
$\mathrm{Tr}\left[\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\right] = 4\left(\eta^{\mu\nu}\eta^{\alpha\beta} - \eta^{\mu\alpha}\eta^{\nu\beta} + \eta^{\mu\beta}\eta^{\alpha\nu}\right)$
$\mathrm{Tr}\left[\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\right] = 4\ i\ \epsilon^{\mu\nu\alpha\beta}$
$\epsilon^{\mu\nu\alpha\beta}\epsilon_{\rho\sigma\alpha\beta} = -2(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\nu_\rho\delta^\mu_\sigma)$

Subject: Research Methodology (PH4203)
End-Semester Examination 2025

Answer Question 1 and any 4 questions from the rest. Full marks: 50

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1. (a) If you are in the DPS or DMS, imagine yourself as Schrödinger and suppose you are writing the paper introducing the Schrödinger equation.
If you are in the DBS or DCS, imagine yourself in the position of Watson and Crick, and suppose you are writing the paper describing the structure of the DNA.
If you are in the DES, imagine yourself as Alfred Wegener, and suppose you are writing the first paper introducing the concept of continental drift in 1912.

Write the abstract of the paper. You have to assume the state of knowledge as existed at the time of writing the paper, but have to use modern convention and stylistics in writing the abstract. (6)

(b) What is the meaning of the term 'falsifiability'? State the *falsifiability criterion* of the following statements.
Statement 1: "The disease AIDS is caused by an agent that is transmitted through body fluids"
Statement 2: "The Earth and the planets were created along with the sun, from the gravitational collapse of a single nebula"
Statement 3: "Chloro-fluoro-carbon compounds are responsible for depletion of ozone layer." (4)

2. (a) The Big Bang theory predicted that there should be a low-temperature radiation in the microwave band—remnant of the primordial explosion. In 1965, the microwave background radiation was discovered. Does the discovery prove the Big Bang theory? Briefly justify your answer. (3)

(b) Suppose you measure three quantities x, y, z by taking 25 measurements each and the results were as follows:

$$\bar{x} = 200,\ s_x = 10, \quad \bar{y} = 50,\ s_y = 10, \quad \bar{z} = 40,\ s_z = 10$$

where the errors in the three quantities are independent and random. Find the quantity $q = x/(y - z)$ with its error bar. (7)

3. (a) Suppose, in a paper a biologist writes that the measured value of a parameter is 5.2±0.05. What information is conveyed by that? The reviewer demands that the error bar should be halved. How should she conduct the experiment to satisfy that demand? (3)

(b) Suppose, in the e/m experiment in Physics lab, you got the following data set:

Accelerating Voltage (volt)	150
Square of Diameter (cm^2)	63, 62.8, 65, 64.2, 66,62, 65.5, 64.8, 66.1, 67

Do you have sufficient evidence to say that the average value of the 'square of diameter' for accelerating voltage 150 is greater than 64? Here 'sufficient evidence' means with 5% significance level. Those who don't know the e/m experiment may simply say A= Accelerating Voltage and B=Square of Diameter, and carry on. (7)

4. (a) Write down the Kirchoff's Current Law equation for the circuit shown below.

(I am just kidding)

The real question is: Do you think the Kirchoff's Current Law will work for a situation like this? If your answer is 'Yes', what is the philosophical basis for the assertion (after all, nobody has really tested it)? If your answer is 'No', what is the philosophical basis for the scepticism? (3)

(b) Newton's theory failed to explain the precession of the perihelion of Mercury. Can Newton's theory still be said to be 'true'? Explain your reasoning. (3)

(c) Draw a box-and-whisker plot for the data given in question 3. (4)

5. (a) Which actions by a scientist are categorized as scientific misconduct? (4)

(b) A new drug is proposed to lower total cholesterol. A randomized controlled trial is designed to evaluate the efficacy of the medication in lowering cholesterol. 52 participants are enrolled in the trial and are randomly assigned to receive either the new drug or a placebo. The participants do not know which treatment they are assigned. Each participant is asked to take the assigned treatment for 6 weeks. At the end of 6 weeks, each patient's total cholesterol level is measured and the sample statistics are as follows:

Treatment	Sample Size	Mean	Standard Deviation
New Drug	25	195.9	28.7
Placebo	27	227.4	30.3

State the null and alternate hypotheses.
Is there statistical evidence of a reduction in mean total cholesterol in patients taking the new drug for 6 weeks as compared to participants taking placebo? (6)

6. (a) Should positivism be accepted as guiding philosophy in science today? Briefly explain the logic behind your opinion. (5)

(b) A survey found that 80 out of 1600 students of IISER-K are left-handed. With what level of confidence can you state that, in general, the proportion of left handed people lies in [0.04, 0.06]? (5)

mechanics

Final Exam PH 4202 Adv. Stat. Phys.

Duration 150 minutes Max Marks 50

1. "Compressible" η^4 model: Consider the following Landau free energy with two coupled order parameters η and u:

$$L = a(T - T_c)\eta^2 + B\eta^4 + \frac{1}{2}u^2 + \lambda\eta^2 u, \quad a, B, \lambda > 0.$$

What is the critical value of the coupling strength λ for the system to have a first-order phase transition?

(4 marks)

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2. Field-driven transition: Consider the Landau free energy

$$L = a(T - T_c)\eta^2 + b\eta^4 - h\eta, \quad a, b, T_c > 0.$$

(a) Argue that there is a field $h-$driven first-order transition for $T < T_c$. The variation of which quantity justifies this claim?

(b) Draw rough plots of (i) the free energy L versus the order parameter η for different values of the field h and (ii) the order parameter η vs. the field h for $T < T_c$ to highlight nature of the phase transition.

(c) Determine the critical magnetic field h_c that defines the limit of metastability (spinodal points).

(d) What happens to the usual temperature induced second-order transition when $h \neq 0$?

(7 marks)

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3. Show that the isothermal susceptibility

$$\chi_T \equiv \tilde{\chi}(k = 0) = \beta \int d^d r\, G(r).$$

Given that at the critical point $G(r) \sim r^{-(d-2)}$ (in the mean field approximation), infer the relationship between the critical exponents ν and γ.

(6 marks)

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4. Given

$$L = -h\eta + a_2 t\eta^2 + a_4\eta^4,$$

determine the ratio

$$\frac{\chi(t = 0_+)}{\chi(t = 0_-)}$$

of the susceptibility $\chi(t)$ as the critical point ($t = \frac{T - T_c}{T_c} = 0$) is approached from above and below.

(4 marks)

RA

5. Give a very rough estimate for the phase transition temperature for the 2D Ising model on a square lattice based on an energy-entropy competition argument.

(5 marks)

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6. Assume that a model equation of state can be written as

$$h \sim aM(t + bM^2 2)^{\theta}; \quad 1 < \theta < 2; \quad a, b > 0$$

near the critical point. Find the exponents β, γ, and δ.

(6 marks)

---

7. Consider a system with continuous symmetry having the free energy

$$\beta L = \int \left(\frac{1}{2}|\nabla\phi|^2 + \frac{r_o}{2}\phi^2 + \frac{u_0}{4}\phi^4\right) d^d r$$

with a two-component order parameter $\vec{\phi}(\vec{x}) = \{\phi_1(\vec{x}), \phi_2(\vec{x})\}$. Working within the Gaussian approximation, show that the transverse fluctuations in the order parameter develop a stiffness in the ordered phase. Show that the corresponding transverse correlation function $\tilde{G}^0_{\perp}(\vec{k})$ has a power law form. Briefly interpret these facts.

(6 marks)

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8. Renormalisation Group: Let the recursion relation for a thermal variable K under a scale change by a factor of 2 be of the form,

$$K' = K^3 + 3K^2(1 - K).$$

(a) Find the non-trivial ($K^* \neq 0, 1$) fixed point K^* for this recursion relation.

(b) Write down the linearized recursion relation. Is the fixed point relevant or irrelevant.

(c) What is value of the critical exponent ν?

(6 marks)

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9. Let the singular part of the magnetic susceptibility χ be given by the following scaling relation

$$\lambda \cdot \chi(t, h) = \lambda^{2a_h} \cdot \chi(\lambda^{a_t} t, \lambda^{a_h} h).$$

Determine the value of the critical exponent γ in terms of a_h and a_t and show that the $\gamma = \gamma'$. Here h is the magnetic field, t the reduced temperature, and γ and γ' are the exponents for the divergence of susceptibility as the temperature is changed to reach the critical point from above and below respectively.

(6 marks)

May 06, 2025

Instructor: Bhavtosh Bansal

PH3203: Advanced Quantum Mechanics

End Semester Examination

28th April 2025

Total Marks: 50 **Time - 2 hour 30 minutes**

Instructions: *Circle the correct option(s) in the question paper. Full credit will be awarded only if all correct options are selected. Partial credit will be given for multiple-select questions based on the number of correct choices marked. However, selecting any incorrect option will result in zero credit.*

Name..... **Roll No**.....

Marks: $(10 \times 5 = 50)$

Q.1) Consider a Hard sphere whose potential is given by -

$$V(r) = \begin{cases} \infty, & r \leq a \\ 0, & r > a \end{cases}$$

We throw in a very low energy wave which scatters off of this potential. Calculate the total cross section σ_T for this scattering.

Q.2) Which of the following options is correct for $\langle k|\hat{x}^3|n\rangle$, where $|n\rangle$, $|k\rangle$ are the eigenstates of the number operator?

(a) $\left(\frac{\hbar}{2m\omega}\right)^{3/2} \begin{pmatrix} 0 & 3 & 0 & \sqrt{3} & \cdots \\ 3 & 0 & 6\sqrt{2} & 0 & \cdots \\ 0 & 6\sqrt{2} & 0 & 9\sqrt{3} & \cdots \\ \sqrt{3} & 0 & 9\sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$ (b) $\left(\frac{\hbar}{2m\omega}\right)^{3/2} \begin{pmatrix} 3 & 0 & 0 & 0 & \cdots \\ 0 & 6\sqrt{2} & \sqrt{3} & 0 & \cdots \\ 0 & \sqrt{3} & 6\sqrt{2} & 0 & \cdots \\ 0 & 0 & 0 & 9\sqrt{3} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$

(c) $\left(\frac{\hbar}{2m\omega}\right)^{3/2} \begin{pmatrix} 0 & 3 & 0 & \sqrt{6} & \cdots \\ 3 & 0 & 6\sqrt{2} & 0 & \cdots \\ 0 & 6\sqrt{2} & 0 & 9\sqrt{3} & \cdots \\ \sqrt{6} & 0 & 9\sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$ (d) $\left(\frac{\hbar}{2m\omega}\right)^{3/2} \begin{pmatrix} 0 & 0 & 0 & 3 & \cdots \\ 0 & 6\sqrt{2} & 6\sqrt{2} & 0 & \cdots \\ 0 & 9\sqrt{3} & 9\sqrt{3} & 0 & \cdots \\ 3 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$

Q.3) Now we move on from simple harmonic oscillator and add an anharmonic perturbation to the hamiltonian -

$$\hat{H} = \hat{H}_0^{\text{SHO}} + \lambda \hat{x}^3$$

We know the expectation of position is zero for S.H.O. But since anharmonic potential is asymmetric about the origin, we expect the expectation of position to be non-zero here. Calculate $\langle \hat{x} \rangle$ in $|n\rangle$ upto first order in λ explicitly and verify -

(a) $\langle \hat{x} \rangle = \frac{3}{2} \frac{E_n^{(0)}}{(m\omega^2)^2}$ (b) $\langle \hat{x} \rangle = \frac{1}{2} \frac{[E_n^{(0)}]^{\frac{3}{2}}}{(m\omega^2)^6}$

(c) $\langle \hat{x} \rangle = \frac{3}{2} \frac{E_n^{(0)}}{(m\omega)^2}$ (d) $\langle \hat{x} \rangle = \frac{5}{2} \frac{[E_n^{(0)}]^2}{(m\omega^2)^2}$

Q.4) The scattering matrix for a potential in 1-D is given by $S = [(a, b), (c, d)]$. If the potential respects the time reversal symmetry then

(a) $a = d$ and $c = d$ (b) $c = b$ and $|a| = |d|$

(c) $\frac{a}{b} = -\frac{d^*}{c^*}$ (d) $a = c$ and $b = d$

Q.5) The scattering matrix for a potential in 1-D is given by $S = [(a, b), (b, a)]$. The transfer matrix corresponding to this S is given by

(a) $M = \left[[\frac{1}{b}, \frac{a*}{b*}], [\frac{a}{b}, \frac{1}{b*}]\right]$ (b) $M = \left[[\frac{1}{b}, \frac{a*}{b*}], [\frac{a}{b}, \frac{1}{b}]\right]$

(c) $M = \left[[\frac{1}{b}, \frac{a}{b}], [\frac{a}{b}, \frac{1}{b}]\right]$ (d) $M = \left[[\frac{a}{b}, \frac{b}{a}], [\frac{b}{a}, \frac{a}{b}]\right]$

Q.6) Consider a time dependent Hamiltonian $\hat{H}(t)$ whose instantaneous eigen-states are given by $|\psi_n\rangle$, $n \in \mathcal{Z}$. Then -

(a) $\langle \psi_m | \frac{\partial}{\partial t} | \psi_n \rangle = \frac{\langle \psi_m | \dot{H} | \psi_n \rangle}{E_n - E_m}$ (b) $\langle \psi_m | \frac{\partial}{\partial t} | \psi_n \rangle = \frac{\langle \psi_m | H | \psi_n \rangle}{i\hbar}$

(c) $\langle \psi_m | \frac{\partial}{\partial t} | \psi_n \rangle = \left\langle \frac{\partial}{\partial t} \psi_m \middle| \psi_n \right\rangle$ (d) $\langle \psi_m | \frac{\partial}{\partial t} | \psi_n \rangle = \frac{|\langle \psi_m | \dot{H} | \psi_n \rangle|^2}{\langle \psi_m | H | \psi \rangle} \frac{i\hbar}{(E_n - E_m)^2}$

Q.7) Consider a 1-D quantum well of initial width $\omega(t = 0)$ with hard wall boundaries. If the wall of the well moves adiabatically to double its size, then calculate

- a) The dynamical phase accumulated by a particle which was prepared in the ground state of the well at t = 0.
- b) Calculate the geometric phase associated with the particle if the well expanded to double its size and then shrink back to its original size adiabatically.

Q.8) The scattering matrix for a potential in 1-D which respectecs time reversel and Parity symmetry can be written as $S = [(r,t),(t,r)]$. In the parity basis (even,odd) of the scattering states, the scattering matrix is given by

(a) $S = [[t-r,0],[0,t+r]]$ (b) $S = [[t+r,0],[0,t-r]]$
(c) $S = [[0,t+r],[0,t-r]]$ (d) $S = [[0,t-r],[0,t+r]]$

Q.9) The action of the permutation operator $\hat{P}_{123}$ and $\hat{P}_{132}$ in terms of exchange operator $\hat{P}_{12}, \hat{P}_{23}, \hat{P}_{13}$ is given by -

(a) $\hat{P}_{12}\hat{P}_{13}$ and $\hat{P}_{13}\hat{P}_{12}\hat{P}_{13}$ (b) $\hat{P}_{12}\hat{P}_{31}\hat{P}_{12}$ and $\hat{P}_{12}\hat{P}_{13}$
(c) $\hat{P}_{12}\hat{P}_{13}$ and $\hat{P}_{13}\hat{P}_{12}\hat{P}_{13}\hat{P}_{12}$ (d) $\hat{P}_{13}\hat{P}_{12}\hat{P}_{13}\hat{P}_{12}$ and $\hat{P}_{12}\hat{P}_{13}$

Q.10) Consider the following three particle state -

$$|\psi_1\rangle = \frac{1}{\sqrt{12}}\left(2\,|a\rangle\,|b\rangle\,|c\rangle + 2\,|b\rangle\,|c\rangle\,|a\rangle - |a\rangle\,|c\rangle\,|b\rangle - |c\rangle\,|b\rangle\,|a\rangle - |c\rangle\,|a\rangle\,|b\rangle - |b\rangle\,|c\rangle\,|a\rangle\right)$$
$$|\psi_2\rangle = \frac{1}{\sqrt{2}}\left(-\,|a\rangle\,|c\rangle\,|b\rangle + |c\rangle\,|b\rangle\,|a\rangle + |c\rangle\,|a\rangle\,|b\rangle - |b\rangle\,|c\rangle\,|a\rangle\right)$$

Where $\{a,b,c\}$ are distinct quantum numbers of the single particle states. Obtain the matrix representation of the permutation operator $\hat{P}_{12}$ and $\hat{P}_{13}$ in the subspace of $|\psi_1\rangle, |\psi_2\rangle$.

Useful Information:

$$a = \frac{1}{\sqrt{2m\omega\hbar}}(m\omega\hat{x} + i\hat{p}), \quad a^\dagger = \frac{1}{\sqrt{2m\omega\hbar}}(m\omega\hat{x} - i\hat{p})$$

$$[\hat{x},\hat{p}] = i\hbar \quad \text{and} \quad [\hat{a},\hat{a}^\dagger] = 1$$

$$f(\theta) = \frac{1}{2ik}\sum_l (2l+1)\left(e^{2i\delta_l - 1}\right)P_l(\cos\theta) \quad \text{where,} \quad P_l(\cos\theta) \quad \text{is the Legendre polynomial.}$$

Intermediate Electricity and Magnetism

End Sem Test

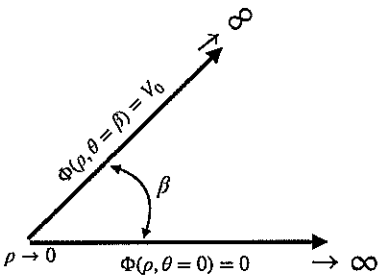
PH3202, Spring 2025 — DPS, IISER Kolkata

Instructor: Dr. Rajesh Kumble Nayak
10:00, 06 May, 2025.
Duration 120 (+ 30) Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Question-1 [20 Marks]



Consider a two-dimensional electrostatic configuration in plane polar coordinates (ρ,θ) where the region of interest is a wedge defined by the angular range $0<\theta<\beta$ and radial range $0<\rho<\infty$. Find the electrostatic potential $\Phi(\rho,\theta)$ satisfies Laplace's equation with following boundary conditions:

$\Phi(\rho,0)=0,$ (the boundary at $\theta=0$ is grounded)
$\Phi(\rho,\beta)=V_0,$ (the boundary at $\theta=\beta$ at constant potential)

• Find the components of electric field. • Find the surface charge density close to the origin.

Question-2 [20 Marks]

Scalar wave equation with source $S(x,t)$ is given by

$$\nabla^2\Psi-\frac{1}{c^2}\frac{\partial^2\Psi}{\partial t^2}=-4\pi S(x,t)\,,$$

reduce to inhomogeneous Helmholtz equation by using the Fourier transformation technique and fine the greens function

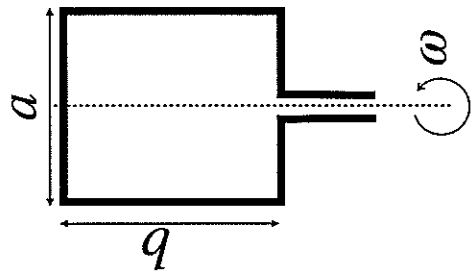
Question-3 [20 Marks]

Consider a thin spherical shell of radius a, carrying a uniform surface charge density σ, is set to rotate at angular velocity ω. Find the vector potential it produces at point $\vec{r}$ outside the sphere.

P.T.O

Rajesh

Question-4 [20 Marks]



Consider a rectangular loop of wire, of width a and length b, rotating with an angular velocity ω an axis as shown in the figure. The loop is lying in a uniform, time dependent magnetic field $B = B_0 \sin \omega t$ perpendicular to the plane of the loop at $t = 0$. Find the $\mathcal{EMF}$ induced in the loop.

Question-5 [20 Marks]

In a region of empty space, the electric field is measured and it is concluded that its value is $\vec{E} = A\,y\,t\,\hat{z}$, where A is a constant, y the coordinate corresponding to a Cartesian reference system, and t the time. Find a possible value for the magnetic field in that region.

Question-6 [20 Marks]

Four point charges are arranged in the xy-plane at the corners of a square of side length $2a$, centered at the origin, as follows:

- $+q$ at $(a, a, 0)$,
- $-q$ at $(-a, a, 0)$,
- $+q$ at $(-a, -a, 0)$,
- $-q$ at $(a, -a, 0)$.

1. Calculate the monopole moment of the system.
2. Compute the dipole moment vector $\vec{p}$, defined by:

$$\vec{p} = \sum_k q_k \vec{r}_k$$

3. Compute the Cartesian components of the quadrupole moment tensor Q_{ij}, defined by:

$$Q_{ij} = \sum_k q_k \left(3x_i^{(k)} x_j^{(k)} - r_k^2 \delta_{ij}\right)$$

4. Use your results from (1)–(3) to find the electrostatic potential $\Phi(\vec{r})$ at a point far from the system ($r \gg a$), retaining only the first three terms (monopole, dipole, quadrupole) in the multipole expansion.
5. Which is preferred coordinate system, Cartesian or Polar and why?

The End

Basic statistical mechanics

PH3201 End Semester Examination

1 May 2025 10:00-12:30AM Full Marks: 50 Attempt all questions.

1. Consider a probability distribution function $f(\epsilon) = 2\sqrt{\frac{\epsilon}{\pi}}e^{-\epsilon}$ where the random variable $\epsilon > 0$.

   Find and sketch the probability distribution function of the transformed random variable $\nu(\epsilon) = \sqrt{\epsilon}$. [8]

2. A Bosonic gas has density of states $g(E) = A E^{\sigma}$ per volume. Empirical observations indicate that $T_c(N) \sim N^{3/7}$.. Assuming the scenario of ideal BEC, estimate σ. [10]

3. For a gas of fermions with spin state $s \equiv \pm 1$, in volume V at temperature T, and in a background magnetic field B

   (a) Write the occupation number of fermions N_+, N_- .

   (b) Find the average magnetisation in the high temperature limit where $f_{3/2}(x) \approx x$.

   (c) Show that magnetic susceptibility in the limit of vanishing B satisfies Curie's law. [12]

4. Write the density of states for acoustic phonons in 2-d lattice given that number of possible polarisations is 2. Hence estimate the maximum allowed phonon frequency ω_D following the argument of Debye. [10]

5. Consider an ideal gas in the isobaric ensemble described by the macro state $M \equiv (N, T, P)$ and micro state $\mu \equiv \{p_i, q_i\}$ with $p_i, q_i \in \mathbb{R}^3$ and $i = 1,2\cdots, N$. Write the partition function and derive the enthalpy. [10]

$$f_{5/2}(z) = \frac{4}{\sqrt{\pi}}\int_0^{\infty} dx\ x^2\ ln[1 + ze^{-x^2}] \qquad f_{3/2} = z\frac{\partial}{\partial z}f_{5/2}(z) \qquad \lambda = \sqrt{\frac{2\ \pi\ \hbar^2}{m\ k_B\ T}} \qquad \sigma = \frac{\pi^2\ k_B^4}{60\ \hbar^3\ c^2}$$

$$g_{5/2}(z) = -\frac{4}{\sqrt{\pi}}\int_0^{\infty} dx\ x^2\ ln[1 - ze^{-x^2}] \qquad g_{3/2} = z\frac{\partial}{\partial z}g_{5/2}(z) \qquad \int_0^{\infty}\frac{x^3\ dx}{e^x - 1} = \frac{\pi^4}{15} \qquad \Gamma(z) = \int_0^{\infty} t^{z-1}e^{-t}dt$$

Mathematical methods of Physics I

Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Endsem Exam, Date: 8 May, 2025 (14:00-16:30)

IISER KOLKATA

Notation: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z) \equiv (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv (\partial_x, \partial_y, \partial_z) \equiv (\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi)$.

1. Solve any two ([4] marks each).
   (a) What type of singularities (if any) possessed by $f(z) = \frac{1}{z-3}$ at $z = 0$ and $z = \infty$?
   (b) Find regions in complex plane where both $f_1(z) = |y| - ix$ and $f_2(z) = |x| + i|y|$ are analytic.
   (c)Find the condition on a, b, c that makes $f(z)$ analytic with $iRe(f(z)) = ax^2 + bxy + cy^2$.

2. Solve any two ([4] marks each).
   (a) Find the particular integral of the differential equation $\frac{dx}{dt} - ax = f(t)$.
   (b) Use Levi-Civita symbols to prove $\det(ABCD) = \det(DABC)$. A, B, C, D are $n \times n$ square matrices.
   (c) Use $z = e^{i\theta}$ to show $4\cos^3(\theta) = \cos(3\theta) + 3\cos(\theta)$.

3. Find Fourier series expansion of $f(x + 2\pi) = f(x) = \begin{cases} 1 + 2x/\pi & -\pi \le x \le 0 \\ 1 - 2x/\pi & 0 < x \le \pi \end{cases}$ [8]

4. Find the Fourier transform of $f(x) = xe^{-ax^2}$, $a > 0$. [8]

5. Find the location(s) on the circle $x^2 + y^2 = 1$ where the magnitude of the gradient of the function $f(x, y) = (x^2 - y^2)e^{-x^2-y^2}$ has its maximum value. What is the value ? [8]

6. Use the matrix method to find the position $x(t)$ of a particle at time t whose acceleration $a(t)$ is changing as $\frac{da}{dt} = 4v + 3a$; $v(t) = \frac{dx}{dt}$ is its velocity and the initial condition is $\{x(0) = 1m, v(0) = 4m/s, a(0) = 16m/s^2\}$. Show that the particle maintains the relation $\frac{x(t)}{x(0)} = \frac{v(t)}{v(0)} = \frac{a(t)}{a(0)}$ at all time $t > 0$. [10]

8/05/2025

Koushik Dutta

PH2202 Thermal Physics
Fall Semester - 2025
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

End-semester Examination Date: 29/04/2025

No clarification regarding the questions will be provided. If any genuine issue is found in any problem, marks will be given appropriately to all.

1. (a) Define first and second order phase transition according to Ehrenfest, and give one example for the each case. **Marks: 3**

   (b) Plot (schematic) Gibbs free energy, entropy, volume and C_p (y-axis) as a function of temperature (x-axis) for a first order phase transition and a second order phase transition. Show two phases clearly. **Marks: 3**

2. The specific volume of vapour is much larger than of the liquid, and vapour behaves like an ideal gas. Assuming that the latent heat of vaporization to be a constant, find out how vapour pressure depends on temperature, and plot it as a function of the temperature. **Marks: 4**

3. (a) Consider an one-dimensional classical simple harmonic oscillator of mass m and angular frequency ω. Find out the total number of microstates having energy $\leq E_0$. Assume each phase-space cell has an area A. **Marks: 3**

   (b) The energy of a quantum mechanical oscillator is given by $E_n = (n + 1/2)\hbar\omega$, where n is the quantum state of the particle. Assume that the oscialltor is usually in states $n \gg 1$, and find out total number of states having energy $\leq E_0$. Demanding that your answer in (b) matches with the answer in (a), find out A. **Marks: 3**

4. The energy of a quantum mechanical oscillator with angular frequency ω is given by $E_n = (n + 1/2)\hbar\omega$, where n $(= 0, 1, 2, 3, 4, ...)$ is the quantum state of the oscillator. Now, consider a system of THREE distinguishable quantum mechanical oscillators, each with angular frequency ω.

   (a) It is found that the total energy (macrostate) of the system is $(7/2)\hbar\omega$. Find out all possible microstates corresponding to the above macrostate. **Marks: 5**

   (b) Assuming that the system is statistical mechanical (It is a bad assumption due to a few number of particles, but let us not worry about it now!), find out the probability (microcanonical) that the first oscillator is in the ground state? **Marks: 2**

5. Consider a system of $N(\gg 1)$ distinguishable spins which can point either UP or DOWN w.r.t some given direction. If a spin points UP, it contributes $+1$ magnetization (in some unit), and if a spin points DOWN, it contributes -1 magnetization (in some unit).

   (a) Find out the total number of microstates Ω for the case when N_+ spins are in the UP direction, and $N_- \equiv N - N_+$ spins are in the DOWN direction. **Marks: 2**

   (b) Show that the entropy of the system is approximately given by

   $$S \simeq k_B \left(N \ln(N) - N_+ \ln(N_+) - N_- \ln(N_-)\right) \ . \tag{1}$$

   Marks: 2

   (b) If M is the total magnetisation, i.e $M = N_+ - N_-$, show that entropy has a *maximum* at $M = 0$. **Marks: 5**

6. The goal of this problem is to find internal energy $U = U(S, V, N)$ for an ideal gas which is described by $U = (3/2)NRT$, and $pV = NRT$. Here, the symbols have their usual meaning.

(a) Using above expressions for an ideal gas, show that

$$\ln U = \frac{2S}{3NR} + f(V, N) \ , \tag{2}$$

where f is an arbitrary function of volume V and total number of particles N. **Marks: 3**

(b) Proceeding further, show the following expression

$$U = g(N)V^{-2/3} \exp[(2S)/(3NR)] \ , \tag{3}$$

where $g(N)$ is an arbitrary function of N. **Marks: 4**

(c) Using extensivity properties of appropriate quantities, find out $g(N)$ to finally get the following expression

$$U(S, V, N) = k \left(\frac{V}{N}\right)^{-2/3} \exp\left(\frac{2S}{3NR}\right) \ , \tag{4}$$

where k is a constant. **Marks: 3**

(d) To check that the answer in part (c) is correct, show $U = (3/2)NRT$, and $pV = NRT$, using the results in part (c). **Marks: 3**

Electricity and magnetism

PH1201 End Semester Examination

Full marks : 40 **Time : 2 hrs 30 minutes**

Name : **Roll No. :**

Attempt any four questions

The final score will be scaled to 50

Q 1) A total charge Q is distributed over a sphere of radius R. The charge density is spherically symmetric and falls off with distance from the center of the sphere r according to the formula $\rho(r) = \rho_0\left(1 - \frac{r}{R}\right)$. An equal but opposite charge $-Q$ is uniformly distributed in a concentric shell between radii R and $2R$.

(a) Determine the constant ρ_0 in terms of Q and R.

(b) Find the electric field everywhere using Gauss law.

(c) Determine the electrostatic potential as a function of r . [2 + 4 + 4]

Q 2) a)(i) Show that the electric field's normal component is discontinuous across a surface carrying a surface charge density σ. Determine the size of this discontinuity.

(ii) On the other hand, show that the tangential component of an electrostatic field is continuous.

b) Find the r dependence of a spherically symmetric static charge density, $\rho(r)$, such that the magnitude of the corresponding electric field is given by $E(r) = A\left(1 - \exp\left(-\frac{r}{a}\right)\right)$ where a and A are constants. [(3 + 2) + 5]

Q 3) a) A particle of mass m and charge q is initially at rest at the origin in a region with uniform electric and magnetic fields given by $\vec{E} = \alpha\hat{k}$ and $\vec{B} = \beta\hat{k}$ where α and β are constants. Determine the position and the velocity of the particle at time t.

b) A stream of particles, each with charge q and mass m , moving along the positive X axis are sent through a velocity selector where the electric and magnetic fields are $E_0\hat{k}$ and $-B_0\hat{j}$, respectively. The velocity selector is designed to block all particles that travel in any direction other than the X axis. The particles which pass through the velocity selector are then sent into a second region where they travel through a uniform electric field $E_1\hat{i}$ for a distance D. Finally the particles are sent through another region carrying a uniform magnetic field $B_1\hat{k}$. What is the radius of the path these charges travel in now? [4 + 6]

Ananda DasGupta

Q 4) a) In class we showed that the magnetic field is a pseudovector - while the electric field is a vector. In order to do this we had assumed - quite naturally, as we thought - that charge is a scalar. However, it turns out that we can just as well assume that charge is a pseudoscalar - so that under parity it obeys $q \mapsto -q$. Show that under this assumption, the electric and magnetic fields must be pseudovector and vector, respectively.

b) Show that even under this assumptions of part (a) above

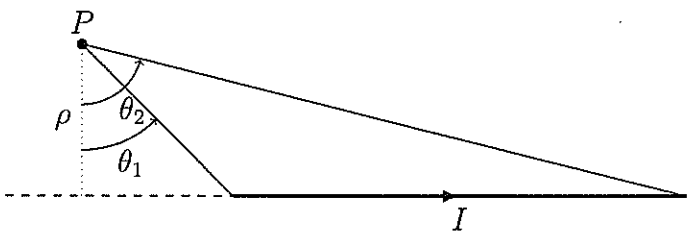
(i) the electric field due to an infinite straight uniformly charged wire would be radial.

(ii) the magnetic field due to an infinite straight current carrying wire would be tangential. $[2 + (4 + 4)]$

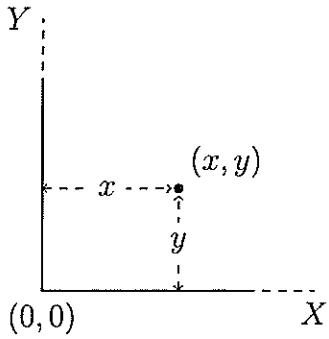
Q 5) We know that the magnetic field caused by a straight segment of wire carrying a current I at a point P is given by

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\sin\theta_2 - \sin\theta_1}{\rho} \hat{\phi}$$

where ρ, θ_1 and θ_2 are indicated in the picture below.



a) An infinite wire is bent so that the two halves make a right angle. The wire carries a current I. We use a coordinate system such that the two halves of the wire lie along the positive X and the Y axes (the bend being at the origin). Determine the magnetic field at a point with coordinates (x, y) in the first quadrant.
Note that the wire extends to infinity along the positive X and Y axes.



b) Determine the magnetic field due to a uniformly wound infinite solenoid with N turns per unit length, carrying a current I, using Ampére's law. $[5 + 5]$

Anand Dev Gupta

Spring End semester 2025 question paper					
Spring 2025 :: Offered Courses (CDS)					
Sl. No.	Course Co	Course Name	Instructor(s)	Submitted	Received
1	CS4201	Information Retrieval and Web Search	Dwaipayan Roy	✓	

Spring End semester 2025 question paper					
Spring 2025 :: Offered Courses (HSS)					
Sl. No.	Course Co	Course Name	Instructor(s)	Submitted	Received
1	HU3201	Introduction to Economics	Tushar Kanti Nandi	✓	
2	HU3202	Introduction to Psychology	Payal Banerjee	✓	

Spring End semester 2025 question paper					
Spring 2025 :: Offered Courses (CESSI)					
Sl. No.	Course Co	Course Name	Instructor(s)	Submitted	Received
1	SS4201	Fluid and Magneto-hydrodynamics	Sudip Kumar Garain	✓	

EndSem Exam - Spring 2025
SS4201: Fluid and Magneto-hydrodynamics

Answer Question 1 and any 4 out of the 5 remaining questions.

Note: Meaning of variables and symbols are same as we used in the class, unless stated otherwise. Clearly mention any assumption that you make while solving any problem.

Time : 150 min Marks: 30

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1. a) Explain how does Faraday rotation act as a depolarizing effect. (2)

b) Explain how do $\mathbf{B}^2/8\pi$ represent the magnetic pressure, whereas, $(\vec{B}.\vec{\nabla})\vec{B}$ represent the magnetic tension. (2)

c) Consider a charged particle being acted on by a uniform force $\vec{F}$ in a direction perpendicular to a uniform magnetic field $\vec{B}$. Write down the equation of motion and sketch the trajectory of the particle clearly explaining your drawing. (2)

d) Discuss how does magnetic mirror work. (2)

e) For dynamical study of compressible flow, one needs to solve the full set of hydrodynamics equations. However, for dynamical study of incompressible flow, one cares about only vorticity equation and doesn't care about continuity or energy equations. Justify this by showing that these two equations are redundant. (2)

2. P_{ij} is given as

$$P_{ij} = nm\frac{\int d^3u w_i w_j f(\vec{x},\vec{u},t)}{\int d^3u f(\vec{x},\vec{u},t)},$$

where, n is particle number density, m is mass of a particle, $\vec{u}$ is total velocity, w's are the components of random velocity and f represents the distribution function. Assuming f to be Maxwell-Boltzmann distribution function as

$$f = n\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-\frac{mw^2}{2kT}},$$

show that

$$P_{ij} = nkT\delta_{ij}$$

(5)

3. a) Astrophysicists used the de Laval nozzle theory to explain the observations of highly supersonic astrophysical jets. Describe how such a nozzle converts a sub-sonic flow to super-sonic speed. (2.5)

b) In Poiseuille's experiment, the kinematic viscosity ν of water is determined using the following formula:

$$\nu = \frac{\pi\ \Delta P}{8Ql} a^4,$$

where, ΔP is the steadily maintained pressure across a pipe of radius a and length l and Q is the mass flux rate of water flowing through the pipe. Derive this formula from Navier-Stokes equation? (2.5)

4. a) Consider the following equation of motion for each species in a two-fluid plasma:

$$m_s n \frac{\partial \vec{v}_s}{\partial t} = -\vec{\nabla} p_s + n q_s (\vec{E} + \frac{\vec{v}_s}{c} \times \vec{B}) - n q_s \eta \vec{j}$$

Here, s is the index representing electron and ion species. By suitably combining these two equations, derive generalized Ohm's law. (2)

b) In order to arrive at the Ohm's law to be used in single fluid MHD, some of the terms are neglected. Discuss the conditions under which these terms are neglected. (1)

c) Finally, using these results, derive the MHD induction equation. (2)

5. Imagine a two-fluid plasma consisting of electrons and protons having masses m_e and m_p respectively. Show that under cold plasma approximation, this plasma undergoes a natural oscillation corresponding to the angular frequency

$$\omega = \omega_p \sqrt{1 + \frac{m_e}{m_p}},$$

where, $\omega_p = \frac{4\pi n_0 e^2}{m_e}$. (You may use following Maxwell's equations in cgs unit: $\nabla \times \vec{B} = \frac{4\pi}{c}\vec{J} + \frac{1}{c}\frac{\partial \vec{E}}{\partial t}$; $\nabla \times \vec{B} = -\frac{1}{c}\frac{\partial \vec{B}}{\partial t}$) (5)

6. a) In Kolmogorov's equilibrium theory, a turbulent flow velocity field is thought of as being made of many eddies of different sizes. Assuming that the smallest size eddies have Reynolds number $R \sim 1$ and that of the largest size as R_L, show that

$$V/v_d \sim R_L^{1/4},$$

where, v_d is the velocity associated to smallest eddies and V is that of largest eddies. (3)

b) When applying Navier-Stokes (N-S) equation in a turbulent situation, the time evolution of the mean velocity is given by a slightly different version than the N-S equation and is known as Reynold's equation. Derive this equation and point out the difference. (2)